

CSC10011 Software Engineering for AI-enabled Systems

Project Assignment 4 (PA4)

Due Date: **10/7/2026**

Total score: 80 points

This is a project assignment which will be submitted by each team.

You have the following documents to be done and submitted. You can write either in **Vietnamese** or **English**.

a – Implementation (5 points)

Your team is required to continue implementing your application's features. The grading for this requirement in this PA is mainly for checking whether your team continues working on implementation or not. The implementation will be graded in PA5 during your presentation. However, your team has to implement key features of your application before performing testing activities required in Section *b* below.

b – Test plan and test cases (30 points)

Software testing is an indispensable part of software development. In this requirement, you are asked to prepare a test plan, design test cases, execute test cases, summarize and report test results. All templates of these documents are provided on Moodle.

Test plan: Fill all sections specified in the test plan template. You need to list the use cases/features to be tested (Section 2 of the Test Plan).

Test cases: You need to write test cases for testing the application. If your application has more than 10 use-cases, write test cases for 10 use cases. You must write **at least 5 test cases** for each use-case. After that, execute the designed test cases for the use-cases that your team has implemented. After each test execution, write the test case status Pass/Fail. Please refer to the test case template provided for more details.

Test report: Generate a test report listing all defects found. If there is a failed test case, there must be a defect that is linked to it. In the **test summary report** sheet, you should include the number of use-cases be tested, number of test cases, number of passed test cases, and number of failed test cases of a use-case.

c – Evaluating ML models (40 points)

In this assignment, you are required to evaluate the ML models implemented and deployed in your application using the Model Evaluation template. You need to show the performance of the models deployed with real data. You need to perform experiments to show whether your models are working or not, by comparing them with baselines or other models using relevant performance metrics.

The evaluation must include descriptions of the following. But if you do not include, you need to provide reasons for not doing so:

- Datasets used for evaluation.
- Approaches, models, methods or some forms of baselines with which your models are compared.
- Performance metrics, such as accuracy and cost.
- Environment setup: machines, tools used, etc.
- Results: report results in forms of tables, figures, diagrams, etc. and descriptions of the results.
- Conclusion: summarize the experimental results and provide conclusions concerning the models used (e.g., whether the models are relevant for the problem, limitations, and future improvements).

c – Weekly report (5 points)

Prepare weekly reports for the weeks starting from the last reported week until the time of submission. Fill out the template with all information required. We expect that the weekly report includes the fine-grained tasks carried out by team members.

Submission Rules

All you students must copy all documents to the directory named PA4-Group[Group ID] and compressed the whole directory to file PA4-Group[GroupId].zip/.rar

For example: PA4-Group01.zip

Grading Criteria

The assignment will be graded according to how well each team member and your team prepare the required documents using the limited information your team has at this time. Criteria include:

- Writing: easy to understand, clear, few errors

- Well formatted texts, diagrams, charts
- The test plan reflects the requirements and the overall project plan.
- Clear sets of test cases with detailed steps and necessary information included.
- Runnable test scripts.
- Automated tests must be done individually and submitted together in one package as detailed above.
- Evaluation of ML models are performed correctly, and results are reported clearly.